

NAME \_\_\_\_\_



GOSFORD HIGH SCHOOL

2016

Higher School Certificate

# MATHEMATICS

## Extension 1

### Assessment Task 1

#### General Instructions

- Reading time 5 minutes
- Working time 60 minutes
- Write using black or blue pen
- Board approved calculators may be used
- For all questions show relevant mathematical reasoning and/or calculations
- Start each question on a new page
- Total marks 42

#### Question 1

Mathematical Induction (10 marks)

a) Prove, using mathematical induction, that

$$5^n + 3 \text{ is divisible by } 4, \text{ for all integers } n \geq 1 \quad (4)$$

b) Prove, by mathematical induction, that

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1) \times 2^n \quad (4)$$

c) **Consider** the statement that

$$\sum_{k=1}^n k^3 = \frac{n^2(n+1)^2}{4}$$

is true for all positive integral values of  $n$ .

Prove that the result is true for  $n = 3$ .

(2)

(NOTE that you are NOT asked to PROVE the result in (c) by MATHEMATICAL INDUCTION)

### Question 2

Polynomials (16 marks)

- a) (i) Find the remainder when  $2x^3 - 5x^2 - 4x + 3$  is divided by  $(x + 2)$  (1)
- (ii) Determine the value of  $p$  for which  $(x - p)$  is a factor of  $A(x)$ ,  
given  $A(x) = x^3 + px^2 - (2p^2 + 12)x + (7p + 10)$  (2)
- b) Neatly sketch  $y = (1 - x)^3(x + 2)$  without the use of calculus,  
(do not attempt to find the turning points) (2)
- c) If  $\alpha, \beta$ , and  $\gamma$  are the roots of the equation  $x^3 - 2x^2 - 3x + 4 = 0$ ,  
determine the value of :
- (i)  $\alpha + \beta + \gamma$  (ii)  $\alpha\beta + \alpha\gamma + \beta\gamma$  (iii)  $\alpha^2 + \beta^2 + \gamma^2$  (4)
- d) Find the value of the constant  $m$  in the equation  $4x^3 + 32x^2 + mx + 60 = 0$ ,  
given that one root is equal to the sum of the other two. (3)
- e) The polynomial  $(x - a)^3 + b$  is equal to zero at  $x = 1$  and when divided by  $x$ ,  
the remainder is  $-7$ .  
Find all possible values of  $a$  and  $b$ . (4)

### Question 3

Parametric Representation (16 marks)

- a) Two distinct points  $P(2ap, ap^2)$  and  $Q(2aq, aq^2)$  lie on the parabola  $x^2 = 4ay$ ,
- (i) show that the gradient of the tangent at  $P$  is  $p$  (1)
- (ii) show that the equation of the tangent to the parabola at  $P$  is  
 $px - y - ap^2 = 0$  (1)
- (iii) show that equation of the chord  $PQ$  is  $(p + q)x - 2y - 2apq = 0$  (2)
- b) A parabola has parametric equations  $x = 4t$ ,  $y = -2t^2$ .
- (i) Find the cartesian equation of this parabola. (1)
- (ii) The normal at any point  $(4t, -2t^2)$ , where  $t \neq 0$ , on the parabola cuts  
the  $y$ -axis at  $G$ .  
Find the equation of this normal and the coordinates of  $G$ . (3)
- (iii) Find the cartesian equation of the locus of the point  $M$ ,  
if  $M$  is the midpoint of  $PG$ . (2)
- c) The point  $P(2ap, ap^2)$  lies on the parabola  $x^2 = 4ay$ , where  $a > 0$ .  
The equation of the normal at  $P$  is  $x + py = 2ap + ap^3$ ,  
**NOTE you do not need to prove this.**
- (i) This normal meets the parabola again at the point  $Q(2aq, aq^2)$ .  
Prove that  $p^2 + pq + 2 = 0$  (3)
- (ii) If the chords  $OP$  and  $OQ$  are perpendicular, where  $O$  is the origin,  
show that  $p^2 = 2$ . (3)

ASSESSMENT TASK 1SOLUTIONSQuestion 1a) Prove true for  $n=1$ 

$$5^n + 3 = 5^1 + 3 \text{ when } n=1$$

$$= 8 \text{ which is divisible by 4}$$

 $\therefore$  true for  $n=1$ Assume true for  $n=k$ 

$$\text{i.e. Assume } 5^k + 3 = 4M \text{ where } M \in \mathbb{J}^+$$

$$\therefore 5^k = 4M - 3$$

Prove true for  $n=k+1$ , if true for  $n=k$ i.e. Prove that  $5^{k+1} + 3$  is divisible by 4.

$$\text{Now } 5^{k+1} + 3 = 5 \times 5^k + 3$$

$$= 5 \times (4M - 3) + 3 \text{ (using assumption)}$$

$$= 20M - 15 + 3$$

$$= 20M - 12$$

$$= 4(5M - 3)$$

which is divisible by 4 since  $(5M - 3)$  is a positive integer. $\therefore$  true for  $n=k+1$ , if true for  $n=k$ .Since true for  $n=1$  and true for  $n=k+1$  if true for  $n=k$  therefore by mathematical induction statement is true for all given 'n'.b) Prove true for  $n=1$ 

$$\text{L.H.S.} = 1 \times 2^{1-1}$$

$$= 1 \times 2^0$$

$$= 1$$

 $\therefore$  true for  $n=1$ 

$$\text{L.H.S.} = 1 + (1-1) \times 2^1$$

$$= 1$$

$$= \text{R.H.S.}$$

Assume true for  $n=k$ 

$$\text{i.e. } 1 \times 2^0 + 2 \times 2^1 + \dots + k \times 2^{k-1} = 1 + (k-1) \times 2^k$$

Prove true for  $n=k+1$ , if true for  $n=k$ 

$$\text{i.e. } 1 \times 2^0 + 2 \times 2^1 + \dots + k \times 2^{k-1} + (k+1) \times 2^k = 1 + k \times 2^{k+1}$$

$$\text{L.H.S.} = 1 + (k-1) \times 2^k + (k+1) \times 2^k$$

$$= 1 + 2^k [(k-1) + (k+1)]$$

$$= 1 + 2^k \times 2k$$

$$= 1 + k \times 2^{k+1}$$

$$= \text{R.H.S.}$$

 $\therefore$  true for  $n=k+1$ , if true for  $n=k$ Since true for  $n=1$ , therefore true for  $n=1+1=2$ Since true for  $n=2$ , therefore true for  $n=2+1=3$ and so on  $\therefore$  By induction true for all given 'n'.

$$\text{c) L.H.S.} = \sum_{k=1}^n k^3$$

$$\text{R.H.S.} = \frac{3^2(3+1)^2}{4}$$

$$= 1^3 + 2^3 + 3^3$$

$$= \frac{9 \times 16}{4}$$

$$= 1 + 8 + 27$$

$$= 36$$

$$= 36$$

$$= \text{L.H.S.}$$

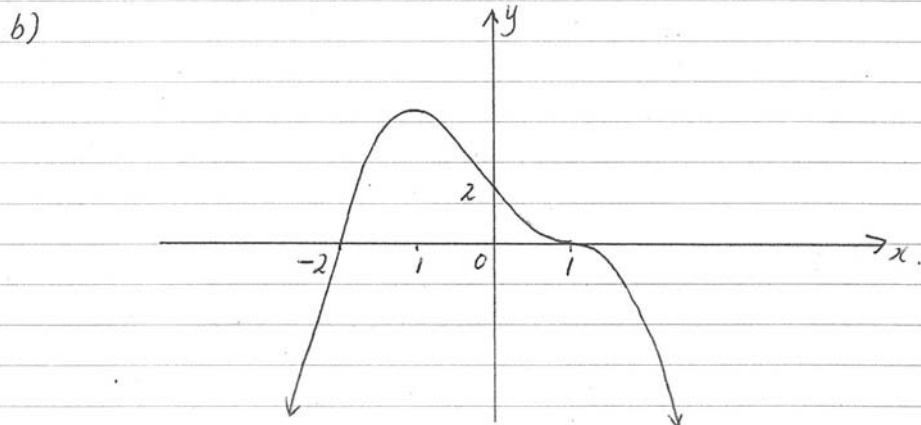
 $\therefore$  true for  $n=3$

## Question 2

## Polynomials

a) i) let  $P(x) = 2x^3 - 5x^2 - 4x + 3$   
 Remainder =  $P(-2)$   
 $= 2(-2)^3 - 5(-2)^2 - 4(-2) + 3$   
 $= -16 - 20 + 8 + 3$   
 $= -25$

ii)  $A(p) = 0$   
 $\therefore (p)^3 + p(p)^2 - (2p^2 + 12) \cdot (p) + (7p + 10) = 0$   
 $p^3 + p^3 - 2p^3 - 12p + 7p + 10 = 0$   
 $-5p + 10 = 0$   
 $p = 2$



c) (i)  $\alpha + \beta + \gamma = -\frac{b}{a}$   
 $= 2$   
 (ii)  $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$   
 $= -3$

(iii)  $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$   
 $= 2^2 - 2(-3)$   
 $= 10$

d) Let the roots be  $\alpha, \beta$  and  $\alpha + \beta$

$$\therefore \alpha + \beta + (\alpha + \beta) = -\frac{32}{4}$$

$$2(\alpha + \beta) = -8$$

$$\alpha + \beta = -4$$

But  $(\alpha + \beta)$  is a root  $\rightarrow -4$  is a root (zero)

$$\therefore 4(-4)^3 + 32(-4)^2 + m(-4) + 60 = 0$$

$$-256 + 256 - 4m + 60 = 0$$

$$4m = 60$$

$$m = 15$$

e) let  $P(x) = (x-a)^3 + b$

$$P(1) = 0$$

$$P(0) = -7$$

$$(1-a)^3 + b = 0$$

$$(0-a)^3 + b = -7$$

$$b = a^3 - 7$$

Solving simultaneously

$$(1-a)^3 + a^3 - 7 = 0$$

$$\therefore 1 - 3a + 3a^2 - a^3 + a^3 - 7 = 0$$

$$3a^2 - 3a - 6 = 0$$

$$a^2 - a - 2 = 0$$

$$(a-2)(a+1) = 0$$

$$a = 2, -1$$

when  $a = 2, b = 1$

when  $a = -1, b = -8$

### Question 3

a) (i)

$$x^2 = 4ay$$

$$y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a}$$

$$= \frac{x}{2a}$$

$$= \frac{2ap}{2a} \text{ at } P$$

$$= p$$

$\therefore$  Gradient of Tangent at  $P$  is  $p$ .

(ii) Equation of Tangent is  $y - ap^2 = p(x - 2ap)$

$$\text{ie } y - ap^2 = px - 2ap^2$$

$$\therefore px - y - ap^2 = 0$$

$$\begin{aligned} \text{(iii) Gradient of chord } PQ &= \frac{ap^2 - aq^2}{2ap - 2aq} \\ &= \frac{a(p-q)(p+q)}{2a(p-q)} \\ &= \frac{p+q}{2} \end{aligned}$$

$$\text{Equation of Chord } PQ \text{ is } y - ap^2 = \frac{(p+q)}{2}(x - 2ap)$$

$$\text{ie } 2y - 2ap^2 = (p+q)x - 2ap(p+q)$$

$$2y - 2ap^2 = (p+q)x - 2ap^2 - 2apq$$

$$(p+q)x - 2y - 2apq = 0$$

$$\text{b) (i) } x = 4t, \quad y = -2t^2$$

$$t = \frac{x}{4} \longrightarrow y = -2\left(\frac{x}{4}\right)^2$$

$$y = -\frac{2x^2}{16}$$

$$y = -\frac{x^2}{8}$$

or

$$x^2 = -8y$$

$$\text{(ii) } y = -\frac{x^2}{8} \longrightarrow \frac{dy}{dx} = \frac{-2x}{8}$$

$$= -\frac{x}{4}$$

$$= -\frac{4t}{4} \text{ at } x = 4t$$

$$= -t$$

$\therefore$  Normal has gradient  $\frac{1}{t}$

$\therefore$  Equation of Normal is  $y + 2t^2 = \frac{1}{t}(x - 4t)$

$$\text{ie } ty + 2t^3 = x - 4t$$

$$x - ty = 4t + 2t^3$$

$$\text{at } t, \quad x = 0 \quad \therefore -ty = 4t + 2t^3$$

$$\therefore y = -4 - 2t^2$$

$$\therefore G \text{ has coordinates } (0, -4 - 2t^2)$$

$$\begin{aligned} M \text{ has coordinates } & \left[ \frac{4t}{2}, \frac{-4 - 2t^2 - 2t^2}{2} \right] \\ & = \left[ 2t, \frac{-4 - 4t^2}{2} \right] \\ & = \left[ 2t, -2 - 2t^2 \right] \end{aligned}$$

$\therefore$  Parametric Equations of M are

$$x = 2t, \quad y = -2 - 2t^2$$

$$t = \frac{x}{2} \rightarrow y = -2 - 2\left(\frac{x}{2}\right)^2$$

$$y = -2 - \frac{x^2}{2}$$

$$2y = -4 - x^2$$

$$\therefore \text{Locus of M is } x^2 = -2y - 4$$

c) Normal has equation  $x + py = 2ap + ap^3$

(i)

$G(2aq, aq^2)$  satisfies

$$2aq + p \cdot aq^2 = 2ap + ap^3$$

$$2q + pq^2 = 2p + p^3$$

$$2q - 2p = p^3 - pq^2$$

$$-2(p - q) = p(p^2 - q^2)$$

$$-2(p - q) = p(p - q)(p + q)$$

$$-2 = p(p + q) \quad \text{since } p \neq q$$

$$-2 = p^2 + pq$$

$$\therefore p^2 + pq + 2 = 0$$

$$(ii) \quad \text{Gradient}(M_{op}) = \frac{ap^2}{2ap} = \frac{p}{2} \quad \text{Gradient}(M_{oq}) = \frac{q}{2}$$

$$M_{op} \times M_{oq} = -1$$

$$\therefore \frac{p}{2} \times \frac{q}{2} = -1$$

$$\therefore pq = -4$$

$$\therefore \text{using (i)} \quad p^2 - 4 + 2 = 0$$

$$\therefore p^2 = 2$$